

Statement of Work (SOW) for Autodesk Revit AI Agent MVP Development

Date: [Insert Date]

BY AND BETWEEN
PKC MANAGEMENT CONSULTING, a partnership firm, having its registered address office at: No.27/7, Dr. Alagappa Road, Purasawalkam, Chennai - 600084 and having GSTIN: 33AACFF9685L1ZW, hereinafter referred to as "PKC" and Represented by Swetha Kochar, Partner
[PKC CONSULTING ENGINEERS PRIVATE LIMITED, A company incorporated under the Companies Act, having its registered office at: {Insert Registered Office Address of PKC} and having GSTIN: [Insert PKC GSTIN], hereinafter referred to as "PKC" and Represented by: Swetha Kochar, Director

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AND

THINKBYTE TECHNOLOGIES PRIVATE LIMITED
A company incorporated under the Companies Act, 2013, having its registered office at: 25-18-42/1, Sampath Nagar, Guntur Andhra Pradesh 522003, India and having GSTIN: 37AAKCT4825H1Z1, hereinafter referred to as "ThinkByte" and Represented by: Vinay Kumar Reddy Kovvuri, Director.

WHEREAS:

- A. PKC is engaged in offering advanced engineering consultancy services, including BIM-enabled architectural and structural modeling across civil and MEP domains. PKC seeks to modernize its workflow by embedding AI-driven automation within Revit.
- B. ThinkByte Technologies specializes in developing agent-first automation solutions, including native tool plugins, multi-agent orchestration, and AI-driven workflow augmentation for enterprise platforms like Revit, IntelliJ, and Figma.
- C. PKC desires to engage ThinkByte as its AI agent development partner to build and deliver an MVP for agents on Autodesk Revit, as detailed in this Statement of Work.

Statement of Work(SOW)

1. Introduction

This Statement of Work outlines the development of an AI-augmented integration platform for Autodesk Revit, purpose-built for PKC Consulting. The solution will be delivered by ThinkByte Technologies (“Provider”) and will establish a secure, cloud-connected ecosystem that enables intelligent automation of architectural workflows directly within Revit.

Background

PKC Consulting is seeking an AI-driven system that can streamline Revit-based workflows such as parametric modeling, iterative design enhancements, and compliance validation. The platform must integrate natively into the Revit environment and support enterprise-grade requirements for security, role-based access, and administrative oversight.

- Commented [1]:** Is the plugin going to be on premise or on cloud?
- Commented [2]:** The Plugin will be inside REVIT. The other two, Revit MCP and AI Agent are outside
- Commented [3]:** Thats right. But the plugin needs to be stored somewhere right? Will that be on Revit Cloud or Revit Onprem. The other question is Revit has Cloud and onprem version - which are we going to build for?
- Commented [4]:** We are building for Revit on prem. But Swetha/Monica will double check.

2. PROJECT SCOPE

2.1 Inclusions

Component	Technical Specifications	Deliverables
Revit Plugin	.NET 6.0/C#, Revit API 2025, WPF UI	- Signed .bundle installer - API communication module - Real-time model sync engine
MCP Server	Python 3.11, FastAPI, Model Context Protocol	- 10+ Revit API tools (walls/doors/windows) - Sandboxed execution environment - Permission control system
Revit Agent	LangGraph 0.1, LlamaIndex 0.10	- Workflow orchestration engine - Tool routing controller

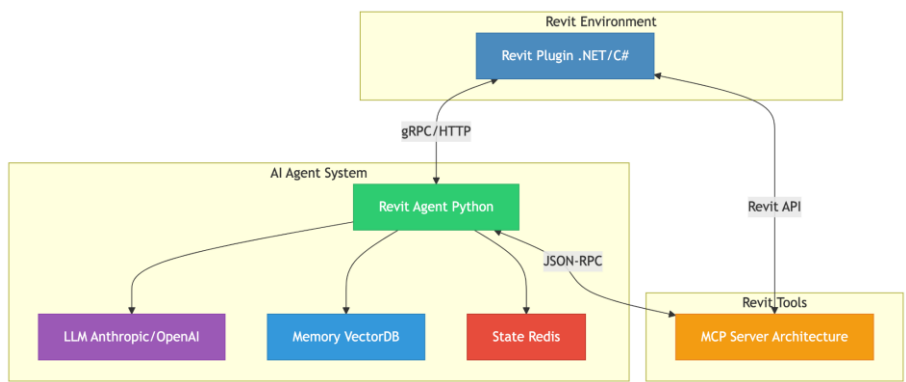
- Commented [5]:** why 10+?
- Commented [6]:** Each action we do on REVIT become a tool through which we do the action. In the initial version, we are planning to target this number
- Commented [7]:** Pls give me some examples of these actions
- Commented [8]:** Like:
Creating a wall
Creating line based elements
Creating point based elements
Creating surface based elements etc
- Commented [9]:** Pending for Monday

2.2 Exclusions

- Custom LLM Models
- On-premises deployment
- Third-party AI model fine-tuning
- Revit version support prior to 2023

3. TECHNICAL ARCHITECTURE

3.1 System Diagram



3.2 Security Protocols

Component	Authentication	Hosting
Revit Plugin	API Key	User
MCP Server	API Key	ThinkByte
Revit AI Agent	API Key	ThinkByte

4. DELIVERY PHASES

Commented [10]: how many firms have this?

Commented [11]: The latest running version is 2025. We can focus the MVP for the latest version, as we will be catering to paid users and they are generally on latest versions.

Commented [12]: ya makes sense. @kocharmonica31@gmail.com do you agree?

Commented [13]: What we build will support 2019-2024

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Phase 1: Core MVP (Weeks 1-2)

Milestone	Acceptance Criteria	Delivery Date
Plugin Framework	Installs in Revit 2025 Registers and executes commands Interacts with the user	Week 1
MCP Tools	Creates walls/doors/windows via API call Retrieves existing information of base elements from Revit	Week 1
Agent v1.0	Processes requests	Week 2
	"Add 3 windows to north wall" command	
	"Add 3 windows to north wall" command	

Commented [14]: will it only execute this much? our expectation is much higher

Commented [15]: The full details are listed below. This is an example

Commented [16]: Ok got it. When you say below, which section are you referring to?

Commented [17]: Interpretable Commands Section (Components Breakdown)

Commented [18]: Pending for Monday

5. HOSTING & INFRASTRUCTURE

Service	Hosting Platform	Specifications	Cost	Additional costs
MCP Server	ThinkByte Platform	-	-	
AI Agent	ThinkByte Platform	LangGraph runtime, LLM proxy	\$50/month (50k node executions per month)	Based on Usage
Total Monthly			\$50/month	Based on usage

Cost Summary:

- Base Hosting Fee: \$50/month (covers MCP + Agent up to 50K nodes)
- Total Monthly Minimum: \$50/month
- 50k node executions per month)

Exclusions:

- 1. Third-party LLM costs (Client must provide OpenAI API key)
- 2. On-premises deployment support
- 3. Training data preparation services
- 4. Revit license costs

6. PAYMENT SCHEDULE

Phase	Amount (INR) Without GST	Trigger	Due	Deliverables Covered
Engagement Fee	₹1,00,000	SOW Execution	Within 3 business days	Platform provisioning, environment setup
MVP Completion	₹3,00,000	Phase 1 Sign-off	Week 2	Revit Plugin + MCP Server + Agent v1.0
Total Project	₹4,00,000			

Financial Notes:

- 1. All cloud costs billed quarterly in advance
- 2. Node execution overage billed monthly in arrears
- 3. 18% GST applicable on all Indian transactions

7. INTELLECTUAL PROPERTY

- 1. Pre-existing IP:
 - ThinkByte retains rights to platform infrastructure, MCP Architecture
 - PKC retains rights to Revit-specific agent and MCP
- 2. Developed IP:
 - PKC owns 100% of Revit Plugin code, Revit Agent Code
- 3. Code Repository:

Commented [19]: Your earlier doc said we retain all IPs including MCPs, API, etc. Pls change it back to that

Commented [20]: In our platform we plan to have 600+ MCP's. The IP for Revit MCP will be retained. Others will stay with the platform.

Commented [21]: Would like to understand this better

Commented [22]: Vinay's final input: All Revit MCPs are PKC IP. All custom built MCPs for PKC will be PKC IP. But other MCPs already available with ThinkByte will be ThinkByte IP. Later when we move out of Think Byte - the cost of moving Revit and other custom MCP will be included in the 2000\$. A few other MCPs may also be included but not if more than that.

Commented [23]: We require the source code for all developments being done for us

- GitHub repository with PKC access

8. SUPPORT & MAINTENANCE

Service	Coverage	Response SLA	Channels
Critical Failures	Business Hours	1 hour	Dedicated hotline
Functionality Issues	Business Hours	4 hours	Teams/Slack
Enhancement Requests	Weekly Reviews	N/A	Google Sheets
Platform Updates	Monthly	N/A	Auto-deployed

9. COMPONENTS BREAKDOWN

Commented [24]: would like you to walk us through this

Commented [25]: Leave this for Monday

1. Revit Plugin (.NET/C#)

Capability	User Impact
Command Execution	Triggers Revit operations safely through queued external events
Connection Management	Maintains real-time WebSocket connection to MCP Server
UI Configuration	Provides settings interface for tool configuration
Command Registry	Manages available Revit commands and their metadata
Security Management	Handles authentication

2. MCP Server (Revit Tools)

Tool Category	Revit Operations
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Element Creation	Creates walls, line-based (beams/pipes), point-based (families), surface-based (floors/ceilings) elements
Element Modification	Changes element properties, applies color overrides
Element Management	Deletes elements
Selection Tools	Gets selected elements, elements in current view
Information Retrieval	Lists loadable families, retrieves view properties

3. AI Agent

Capability	Function	Technical Implementation
MultiAgent Workflow Orchestration	Routes tasks between components	LangGraph state machines
Tool Selection	Chooses appropriate MCP tools	LLM-based tool classification
Error Handling	Recovers from failed operations	Fallback strategies and retry mechanisms
UserIntent Parsing	Translates natural language to commands	Prompt engineering with LLMs

Key Constraints

- 1. Element Limitations:
 - Only supports all/floor/ceiling creation (no structural analysis)
 - Family creation limited to point-based placement
 - No MEP system creation tools
- 2. View Operations:
 - Only current view elements accessible
 - No view creation/management tools
- 3. Data Scope:
 - Can access element properties but not create new parameters

Commented [26]: what does this mean?

Commented [27]: Families are objects created in revit like walls, doors etc.

There are different placement methods like point based, edge based etc. Initially we are aiming for point based, can be re-looked at.

Commented [28]: @kocharmonica31@gmail.com is this ok?

Commented [29]: Lets look on Monday

Commented [30]: dont understand what this means

Commented [31]: View refers to ongoing project. Revit allows changes only to the current active view.

Commented [32]: Makes sense. What does no view creation/management tools mean? @kocharmonica31@gmail.com you may know this better

Commented [33]: Like creation of a new project/ making changes to inactive projects

Commented [34]: Need to discuss on Monday. New viewport creation will be required

- Schedule/data extraction not included

Commented [35]: dont understand

Commented [36]: Schedule means creation of a workflow. The schedule cannot run without the file (The workflow file need to be created in the local plugin to run - similar to demo shown). This requires automation of workflow file.

Commented [37]: Seems like a word is missing. Pls check

Commented [38]: Will relook on Monday

Interpretable Commands:

Valid command patterns

- "Add 3 windows to the east wall"
- "Change all doors to FamilyType 'ACME_Door_102'"
- "Delete red walls in Level 2"

Unsupported commands

- "Redesign the staircase" # No generative design
- "Optimize structural loads" # No engineering analysis
- "Create HVAC system" # No MEP tools

Accuracy Boundaries:

- 90% accuracy on single-element commands
- 75% accuracy on multi-element operations
- Requires element tags/labels for spatial references

10. TEAM STRUCTURE & MEETING CADENCE

ThinkByte Project Team

Role	Team Member	Responsibilities	Availability
Revit Developer	Yash Mori	- Plugin development - Revit API integration - Debugging	Daily
AI Developer	Anupam &	- LangGraph workflows - Tool integration - Agent logic	Daily

Technical Lead	Abhilash	- Architecture oversight - Code reviews - Risk mitigation	Every 3 days
Account Manager	Vamsi, Vinay	- Client communication - Timeline tracking - Requirement validation	Every 3 days

Meeting Schedule (First 2 Weeks)

1. Daily Standups (Mon-Fri)

- Time: 9:30 AM IST (15 mins)
- Attendees: Yash/Anupam + Client Rep
- Format:
 - Yesterday's progress
 - Today's priorities
 - Blockers requiring client input

Tools: Google Meet, Shared Jira Board

2. Bi-Weekly Demos (Every 3 Days)

- Day 3, 6, 9, 12 of project
- Attendees: Full ThinkByte team + Client Team
- Agenda:
 - Live demo of completed features
 - Technical deep dive (Abhilash)
 - Feedback incorporation plan
 - Timeline adjustment review
- Duration: 45 minutes

11.PLATFORM MIGRATION OPTION

Service: LangGraph + MCP Migration Package

Price: \$2,000 (one-time fee)

What's Included

- 1. LangGraph Migration
 - Portable state machine core (Python 3.11)
 - Pre-configured workflow executor
 - Custom node registry template
- 2. Revit MCP Tool
 - All registered tools (TS → Python conversion)
 - Tool validation harness
 - Revit API compatibility layer
- 3. Integration Components
 - FastAPI server with /execute endpoint
- 4. Documentation
 - Deployment guide for AWS
 - Security hardening checklist

What's NOT Included

Capability	ThinkByte Platform Feature	Migration Package
Workflow Tracing	Visual execution tree with timestamps	✗ Not included
Prompt Management	Version-controlled prompt templates	✗ Not included
Auto-Scaling	Kubernetes HPA configuration	✗ Manual setup required
LLM Proxy	Unified API gateway (OpenAI/Azure/Anthropic)	✗ Client must implement
Tool Marketplace	Central registry with UI	✗ Basic tool loader only

Commented [39]: dont understand whats written here. basically, everything built has to migrate. will that happen?

Commented [40]: Yes. Everything custom built for the project will be wrapped in usable layer and transferred

Commented [41]: Got it.

Commented [42]: Monday

Error Diagnostics	AI-assisted failure analysis	 Standard Python logging
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12. SIGNATORIES

By signing below, both parties agree to the terms and conditions outlined in this Statement of Work.

For PKC CONSULTING :

Name: Swetha Kochar
Title: Director
Signature: _____
Date: _____

For ThinkByte Technologies Pvt Ltd :

Name: Vinay Kumar Reddy Kovvuri
Title: Director
Signature: _____
Date: _____
